

Ragul Pazhamalai

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PROFESSIONAL SUMMARY

Data Analytics Professional with 4 years of experience delivering business intelligence solutions using Power BI, SQL, and Python. Microsoft PL-300 Certified with expertise in dashboard development, data modeling, ETL, DAX, and workflow automation. Recently expanded into Generative AI by building Retrieval-Augmented Generation (RAG) applications using LangChain, OpenAI, and vector databases, with a strong interest in AI Engineer and Analytics Engineer roles.

TECHNICAL SKILLS

Power BI & Analytics

Power BI • Power BI Service • DAX • Power Query • Data Modeling • Row-Level Security (RLS)

Programming

Python • SQL

AI & Machine Learning

LangChain • Retrieval-Augmented Generation (RAG) • OpenAI API • Prompt Engineering • chroma • Scikit-learn • Pandas • NumPy

Core Concepts

ETL • Data Visualization • Machine Learning • Generative AI

PROFESSIONAL EXPERIENCE

Senior Data Analyst, Randstad

09/2023 – Present | Chennai

- Managed and maintained 180+ Power BI reports, ensuring performance optimization, timely enhancements, and high report availability.
- Developed interactive dashboards and KPI reports to support business decision-making.
- Designed scalable data models using DAX and Power Query.
- Performed ETL processes using SQL and Power Query to prepare clean and reliable datasets.
- Worked closely with business stakeholders to gather reporting requirements and deliver analytical insights.
- Supported report governance, security, and performance improvements across enterprise reporting solutions.

Client: Caterpillar

Data Analyst, Onward Technologies

07/2022 – 09/2023 | Chennai

- Built Power Apps solutions to digitize manual business processes and improve operational efficiency.
- Developed internal applications including Leave/WFH Management and ECR Request systems with approval workflows.
- Performed data analysis using SQL and Excel to identify trends and generate business insights.
- Collaborated with cross-functional teams to automate reporting and improve process efficiency.

Client: Caterpillar

KEY PROJECTS

Drawing Quality & Root Cause Analysis Dashboard (Power BI)

- Developed an end-to-end Power BI solution to monitor the complete engineering drawing lifecycle from detailing to manufacturing release.
- Designed a Snowflake Schema by creating relationships between fact and dimension tables.
- Developed KPIs including cycle time, process velocity, and error rate to monitor engineering performance.
- Performed root cause analysis to identify process bottlenecks affecting manufacturing timelines.
- Implemented conditional formatting to highlight projects exceeding target turnaround time.
- Configured Row-Level Security (RLS) with four access roles for secure project-level visibility.
- Delivered dashboards to 1500+ users, improving engineering process visibility across multiple teams.
- Reduced drawing release delays by approximately 15–20%.

AI-Powered PDF Question Answering System (RAG)

- Developed an AI-powered document question answering application using Retrieval-Augmented Generation (RAG).
- Built an end-to-end pipeline using LangChain, document chunking, embeddings, and FAISS Vector Database.
- Integrated the OpenAI API to generate accurate, context-aware responses from uploaded PDF documents.
- Implemented semantic search for efficient document retrieval and improved answer relevance.
- Developed an interactive web application using Streamlit, allowing users to upload PDFs and ask natural language questions.
- Applied prompt engineering techniques to improve response quality and reduce hallucinations.
- Built using Python, LangChain, OpenAI API, FAISS, Streamlit, and PyPDFLoader.

CERTIFICATION

Microsoft Certified: Power BI Data Analyst Associate (PL-300)

EDUCATION

SA Engineering College, Chennai, *Bachelor of Engineering – Mechanical Engineering*

2014 – 2018